

### 03 practice examples on where, having, group by, order by

```
mysql> USE practice_db;
Database changed
mysql> CREATE TABLE orders (
    ->     ord_no INT PRIMARY KEY,
    ->     purch_amt DECIMAL(10,2),
    ->     ord_date DATE,
    ->     customer_id INT,
    ->     salesman_id INT
    -> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> INSERT INTO orders VALUES
    -> (70001,150.50,'2012-10-05',3005,5002),
    -> (70009,270.65,'2012-09-10',3001,5005),
    -> (70002,65.26,'2012-10-05',3002,5001),
    -> (70004,110.50,'2012-08-17',3009,5003),
    -> (70007,948.50,'2012-09-10',3005,5002),
    -> (70005,2400.60,'2012-07-27',3007,5001),
    -> (70008,5760.00,'2012-09-10',3002,5001),
    -> (70010,1983.43,'2012-10-10',3004,5006),
    -> (70003,2480.40,'2012-10-10',3009,5003),
    -> (70012,250.45,'2012-06-27',3008,5002),
    -> (70011,75.29,'2012-08-17',3003,5007),
    -> (70013,3045.60,'2012-04-25',3002,5001);
Query OK, 12 rows affected (0.04 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT *
-> FROM orders
-> WHERE purch_amt > 2000;
```

| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
|--------|-----------|------------|-------------|-------------|
| 70003  | 2480.40   | 2012-10-10 | 3009        | 5003        |
| 70005  | 2400.60   | 2012-07-27 | 3007        | 5001        |
| 70008  | 5760.00   | 2012-09-10 | 3002        | 5001        |
| 70013  | 3045.60   | 2012-04-25 | 3002        | 5001        |

4 rows in set (0.01 sec)

```
mysql> SELECT *
-> FROM orders
-> WHERE ord_date = '2012-09-10';
```

| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
|--------|-----------|------------|-------------|-------------|
| 70007  | 948.50    | 2012-09-10 | 3005        | 5002        |
| 70008  | 5760.00   | 2012-09-10 | 3002        | 5001        |
| 70009  | 270.65    | 2012-09-10 | 3001        | 5005        |

3 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM orders
-> WHERE salesman_id = 5001;
```

| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
|--------|-----------|------------|-------------|-------------|
| 70002  | 65.26     | 2012-10-05 | 3002        | 5001        |
| 70005  | 2400.60   | 2012-07-27 | 3007        | 5001        |
| 70008  | 5760.00   | 2012-09-10 | 3002        | 5001        |
| 70013  | 3045.60   | 2012-04-25 | 3002        | 5001        |

4 rows in set (0.00 sec)

```
mysql> SELECT *
```

```
mysql> SELECT *
-> FROM orders
-> ORDER BY purch_amt DESC;
```

| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
|--------|-----------|------------|-------------|-------------|
| 70008  | 5760.00   | 2012-09-10 | 3002        | 5001        |
| 70013  | 3045.60   | 2012-04-25 | 3002        | 5001        |
| 70003  | 2480.40   | 2012-10-10 | 3009        | 5003        |
| 70005  | 2400.60   | 2012-07-27 | 3007        | 5001        |
| 70010  | 1983.43   | 2012-10-10 | 3004        | 5006        |
| 70007  | 948.50    | 2012-09-10 | 3005        | 5002        |
| 70009  | 270.65    | 2012-09-10 | 3001        | 5005        |
| 70012  | 250.45    | 2012-06-27 | 3008        | 5002        |
| 70001  | 150.50    | 2012-10-05 | 3005        | 5002        |
| 70004  | 110.50    | 2012-08-17 | 3009        | 5003        |
| 70011  | 75.29     | 2012-08-17 | 3003        | 5007        |
| 70002  | 65.26     | 2012-10-05 | 3002        | 5001        |

12 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM orders
-> ORDER BY ord_date;
```

| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
|--------|-----------|------------|-------------|-------------|
| 70013  | 3045.60   | 2012-04-25 | 3002        | 5001        |
| 70012  | 250.45    | 2012-06-27 | 3008        | 5002        |
| 70005  | 2400.60   | 2012-07-27 | 3007        | 5001        |
| 70004  | 110.50    | 2012-08-17 | 3009        | 5003        |
| 70011  | 75.29     | 2012-08-17 | 3003        | 5007        |
| 70007  | 948.50    | 2012-09-10 | 3005        | 5002        |
| 70008  | 5760.00   | 2012-09-10 | 3002        | 5001        |
| 70009  | 270.65    | 2012-09-10 | 3001        | 5005        |
| 70001  | 150.50    | 2012-10-05 | 3005        | 5002        |
| 70002  | 65.26     | 2012-10-05 | 3002        | 5001        |
| 70003  | 2480.40   | 2012-10-10 | 3009        | 5003        |
| 70010  | 1983.43   | 2012-10-10 | 3004        | 5006        |

12 rows in set (0.00 sec)

```
mysql> SELECT SUM(purch_amt) AS total_revenue  
-> FROM orders;
```

```
+-----+  
| total_revenue |  
+-----+  
|      17541.18 |  
+-----+  
1 row in set (0.01 sec)
```

```
mysql> SELECT AVG(purch_amt) AS average_order  
-> FROM orders;
```

```
+-----+  
| average_order |  
+-----+  
|    1461.765000 |  
+-----+  
1 row in set (0.00 sec)
```

```
mysql> SELECT MAX(purch_amt) AS highest_order  
-> FROM orders;
```

```
+-----+  
| highest_order |  
+-----+  
|         5760.00 |  
+-----+  
1 row in set (0.00 sec)
```

```
mysql> SELECT COUNT(*) AS total_orders
-> FROM orders;
+-----+
| total_orders |
+-----+
|          12 |
+-----+
1 row in set (0.01 sec)
```

```
mysql> SELECT MIN(purch_amt) AS lowest_order
-> FROM orders;
+-----+
| lowest_order |
+-----+
|         65.26 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT salesman_id,
->         SUM(purch_amt) AS total_sales
-> FROM orders
-> GROUP BY salesman_id;
+-----+-----+
| salesman_id | total_sales |
+-----+-----+
|          5002 |      1349.45 |
|          5001 |     11271.46 |
|          5003 |      2590.90 |
|          5005 |       270.65 |
|          5006 |      1983.43 |
|          5007 |        75.29 |
+-----+-----+
6 rows in set (0.01 sec)
```

```
mysql> SELECT customer_id,
->          SUM(purch_amt) AS total_purchase
-> FROM orders
-> GROUP BY customer_id;
```

| customer_id | total_purchase |
|-------------|----------------|
| 3005        | 1099.00        |
| 3002        | 8870.86        |
| 3009        | 2590.90        |
| 3007        | 2400.60        |
| 3001        | 270.65         |
| 3004        | 1983.43        |
| 3003        | 75.29          |
| 3008        | 250.45         |

8 rows in set (0.00 sec)

```
mysql> SELECT salesman_id,
->          SUM(purch_amt) AS total_sales
-> FROM orders
-> GROUP BY salesman_id
-> HAVING SUM(purch_amt) > 3000;
```

| salesman_id | total_sales |
|-------------|-------------|
| 5001        | 11271.46    |

1 row in set (0.00 sec)

```
mysql> SELECT customer_id,
->          SUM(purch_amt) AS total_purchase
-> FROM orders
-> GROUP BY customer_id
-> HAVING SUM(purch_amt) > 2500;
```

| customer_id | total_purchase |
|-------------|----------------|
| 3002        | 8870.86        |
| 3009        | 2590.90        |

2 rows in set (0.00 sec)

```
mysql> SELECT customer_id,
->          COUNT(*) AS total_orders
-> FROM orders
-> GROUP BY customer_id
-> HAVING COUNT(*) > 1;
```

| customer_id | total_orders |
|-------------|--------------|
| 3005        | 2            |
| 3002        | 3            |
| 3009        | 2            |

3 rows in set (0.00 sec)

```
mysql> SELECT customer_id,
->          SUM(purch_amt) AS total_purchase
-> FROM orders
-> GROUP BY customer_id
-> HAVING SUM(purch_amt) > 1000
-> ORDER BY total_purchase DESC;
```

| customer_id | total_purchase |
|-------------|----------------|
| 3002        | 8870.86        |
| 3009        | 2590.90        |
| 3007        | 2400.60        |
| 3004        | 1983.43        |
| 3005        | 1099.00        |

5 rows in set (0.00 sec)

```
mysql> SELECT customer_id,
->          MAX(purch_amt) AS max_purchase
-> FROM orders
-> GROUP BY customer_id
-> HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
```

| customer_id | max_purchase |
|-------------|--------------|
| 3002        | 5760.00      |
| 3009        | 2480.40      |
| 3007        | 2400.60      |

3 rows in set (0.00 sec)



```
mysql> SELECT salesman_id,
->          COUNT(*) AS total_orders
-> FROM orders
-> GROUP BY salesman_id
-> HAVING COUNT(*) >= 2;
```

| salesman_id | total_orders |
|-------------|--------------|
| 5002        | 3            |
| 5001        | 4            |
| 5003        | 2            |

3 rows in set (0.00 sec)

```
mysql> SELECT ord_date,
->          MAX(purch_amt) AS highest_purchase
-> FROM orders
-> GROUP BY ord_date
-> HAVING MAX(purch_amt) > 2000;
```

| ord_date   | highest_purchase |
|------------|------------------|
| 2012-10-10 | 2480.40          |
| 2012-07-27 | 2400.60          |
| 2012-09-10 | 5760.00          |
| 2012-04-25 | 3045.60          |

4 rows in set (0.00 sec)